Part 1. Foundations

Part 1 consists of five chapters that teach the most fundamental aspects of deep reinforcement learning. After reading part 1, you'll be able to understand the chapters in part 2 in any order.

Chapter 1 begins with a high-level introduction to deep reinforcement learning, explaining its main concepts and its utility. In chapter 2 we'll start building practical projects that illustrate the basic ideas of reinforcement learning. In chapter 3 we'll implement a deep Q-network—the same kind of algorithm that DeepMind famously used to play Atari games at superhuman levels.

Chapters 4 and 5 round out the most common reinforcement learning algorithms, namely policy gradient methods and actor-critic methods. We'll look at the pros and cons of these approaches compared to deep Q-networks.